Computer Science and Engineering Faculty

Syllabus

Subject	**Systems and Signals**			
	Type	**Semester**	**ECTS**	**Code**
	OBLIGATIVE (O)	4	5	
Course Lecturer	Assoc. Prof. Dr Armend Ymeri			
Aims and Objectives	The course aims to provide basic knowledge about signals and their classification; Signal comparison; Signal flow in time; Continuous and discrete signals. Periodic and aperiodic signals. Exponential and trigonometric Fourier series, Fourier transforms and their properties; Interlacing. Energy and power of signals; Systems, their classification and analysis in time and frequency; Amplitude, frequency and phase modulations; Sampling and quantization; Discrete signal processing; Systems analysis and Laplace transforms; Inverse Laplace transforms; Representation of Systems through Block diagrams; Z - Transforms and their application; Inverse Z-Transform; Properties of Z-Transforms The objective of the course is for the student to be introduced to the basic methods for analysis and synthesis of signals and systems and to gain skills in their application, through solving problems and performing relevant simulations. Students will have knowledge in the mathematical and physical interpretation of the nature of signals and their processing within the respective systems. With the inclusion of the laboratory part, they will have enough knowledge to work with real systems in the industry by performing digital computer simulations using MATLAB.			
Learning Outcomes	After completing this course, the student will be able to: • Understand the basic concepts for signals and systems, in continuous and discrete time, as well as to determine their properties; • Apply the basic methods of analysis of signals and systems in the time domain; • Interpret the basic methods of analysis of signals and systems in the frequency domain, through Furies transformations, both in continuous and discrete time; • Understand the basic concepts for filtering, sampling and for the basic ways of modulating signals; • Apply the methods of analysis of signals and continuous time systems in the complex frequency domain through the Laplace transform;			
Course Content	**Course Plan**			**Week**
	Presentation of the Syllabus			1
	Introduction to Systems and Signals			2
	Classification of Signals			3
	Basic Continuous Time Signals and Basic Discrete Signals, Basic Signal Operations			4
	Systems and Classification of Systems			5
	Description of Discrete, Linear, and Displacement-invariant Systems through Impulse Response and Output Determination through Convolution Sum.			6
	Description of Continuous-Time, Linear, and Displacement-Invariant Systems through Impulse Response and Determination of Output through the Weave Integral.			7
	Description of Continuous Periodic Signals through the Fourier series. Description of Continuous Aperiodic signals through Fourier Transformation and Properties of Fourier transformation.			8
	Transformation applications in Communication Systems;			9
	Fourie Transformation Application. Filters and their Classification;			10
	Discrete Fourier Transform;			11
	Laplace Transformation and its Properties. Inverse Laplace Transformation.			12
	Laplace Transformation and its Applications;			13
	Z-Transformation and its Applications,.			14
	Exam Preparation			15

Teaching/Learning Methods	**Activity**		**Weight (%)**
	1. Lectures		70
	2. Numerical Exercises/MATLAB		30
Assessment Methods	**Assessment Activity**		**Weight (%)**
	1. Participation in Lectures		5
	2. Participation in Exercises		10
	3. Design Task/MATLAB		20
	4. Final Exam		60
Course resources	**Resources**		**Number**
	1. Class		1
	2. Moodle		1
	3. Projector		1
ECTS Workload	**Activity**	**Weekly hrs**	**Total workload**
	1. Lectures	1-15	30
	2. Numerical Exercises/MATLAB	1/15	15
	3. Design Task/MATLAB		15
	4. Individual Activity		63
	5. Final Exam		2
Literature/References	• Material and printed notes for lectures and exercises (authorized by the professor and the assistant). • Hwei P. Hsu, "Schaum's Outline of Theory and Problems of Signals and Systems, McGraw-Hill, 2014. • H. Muçostepa, B.Kamo, V.Koliçi, I.Shinko; "Teoria e sinjaleve " (Literaturë e detyruar) ,Tetor, 2016. • Alex Palamides, Anastasia Veloni, "Signals and Systems Laboratory with MATLAB", CRC Press, 2011. • Luis F Chaparro ,,"Signals and Systems Using MATLAB", Academic Press, 2014 • Edward W Kamen "Fundamentals of Signals and Systems Using MATLAB" Prentice Hall, 1997 • B P Lathi ,,"Linear Systems and Signals", 2 nd edition, Oxford University Press		
Contact	Prof. Asoc. Dr. Armend Ymeri e-mail: armend.ymri@ubt-uni.net		